

# SYSTEMS ENGINEERING

## Systems Analysis & Design

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2025-III



# Outline

- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



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# What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance of complex systems**.
- This **discipline** is based on interdisciplinary fields, such as **computer engineering**, **civil engineering**, **software engineering**, **mechanical engineering**, **electrical engineering**, **organizational studies**, **project management**, and others.
- **Systems Engineering** is a holistic approach to engineering that focuses on how to **design and manage complex systems** over their lifecycle.

CGR

software

Physical



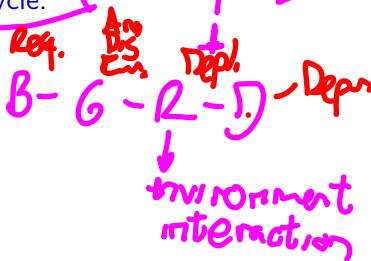
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# Systems Engineer as a Professional

- A **Systems Engineer** is a professional who is responsible for designing, implementing, and maintaining complex systems.

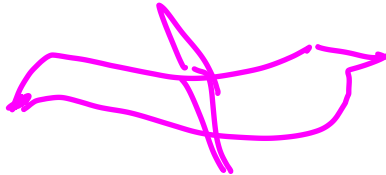
architect  
software  
error-fix



# Systems Engineer as a Professional

- A **Systems Engineer** is a professional who is responsible for designing, implementing, and maintaining complex systems.
- A **Systems Engineer** must have a broad understanding of engineering, mathematics, science, and technology.
- A Systems Engineer must be able to analyze and solve problems in a systematic and holistic way.

Human  
AI Pin



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Yesterday  
 12-6  
 10 problems  
 english

Programming  
 Contest

Google

⇒



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# Human Activities

random

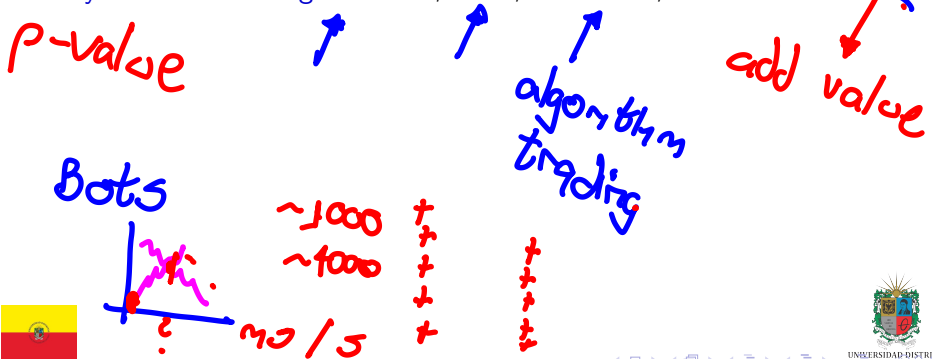
- **Human activities** are complex systems that involve multiple components, interactions, and feedback loops.
- Systems Engineering can be applied to understand, analyze, and improve human activities.
- Systems Engineering can help us design and manage complex systems such as organizations, cities, economies, and societies.

variability



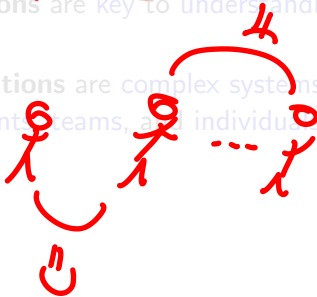
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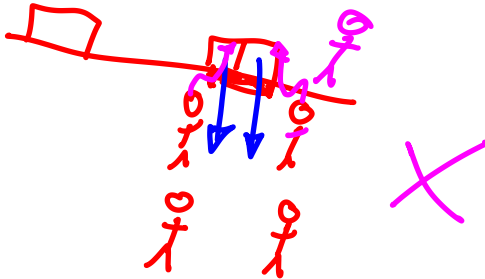
# Human Activities: Complexity

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- **Interactions** are key to understanding and improving human activities.
- **Organizations** are complex systems that involve multiple departmental teams, and individuals.



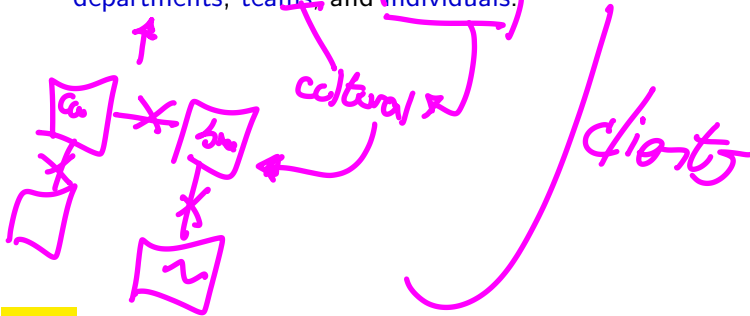
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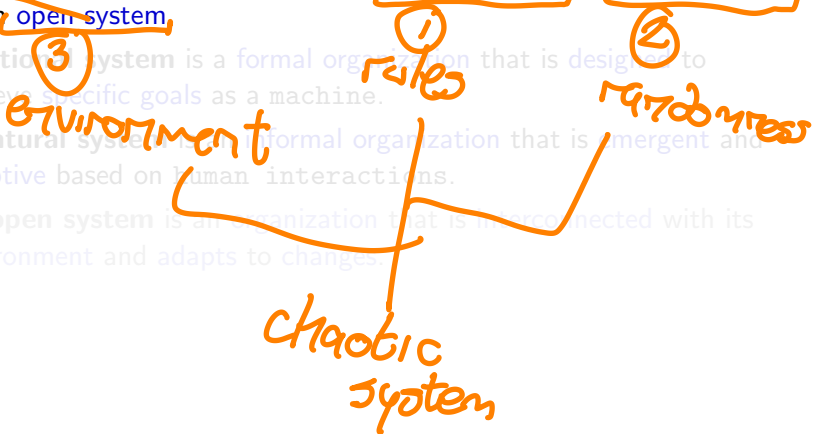
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# Organizations as Systems

- **Organizations** can be viewed as: a rational system, a natural system, or an open system.

- A rational system is a formal organization that is designed to achieve specific goals as a machine.
- A natural system is an informal organization that is emergent and adaptive based on human interactions.
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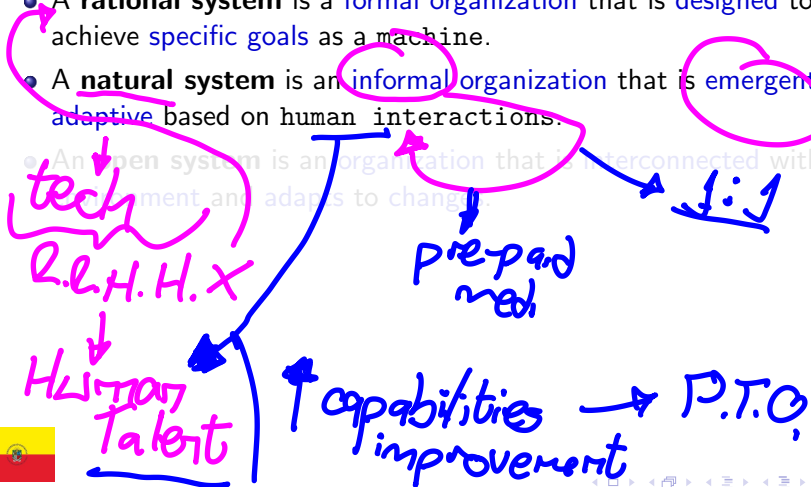
business model

mission  
vision  
goal  
methodology  
internal rules



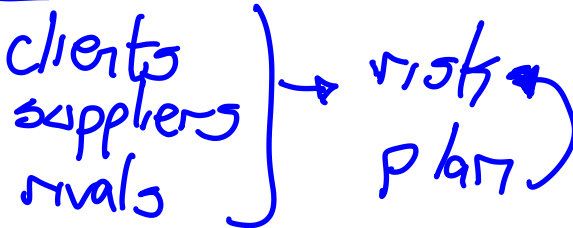
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# Synergy

remote

- **Synergy** is a simple but powerful concept: the whole is greater than the sum of its parts. *Team Football*
- It means the interactions could boost the capabilities of the parts of the system. Also, it lets both understand emergent behaviors and define improvements in systems.

time + simulation

One of the main concepts is the time of computation. Based on graphs, you can define a computational machine.

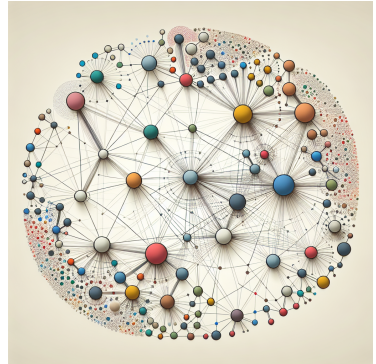


Figure: Prompt: Define a draw of clusters in social networks.



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- It means the **interactions** could **boost the capabilities** of the parts of the **system**. Also, it lets both understand **emergent behaviors** and define **improvements** in **systems**.
- One of the main concepts is the **theory of computation**. Based on **graphs**, you could define a **computational machine**.

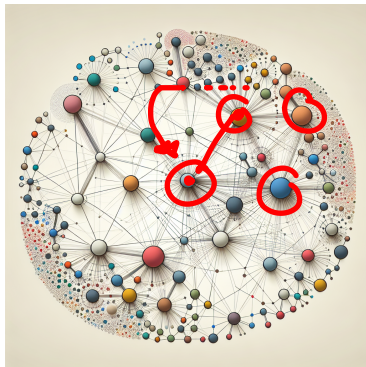


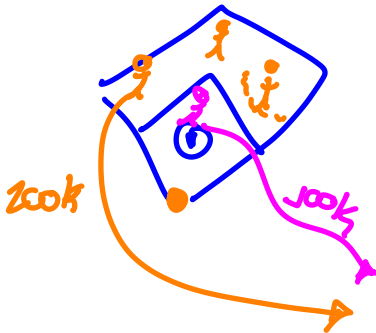
Figure: Prompt: Define a draw of clusters in social networks.

1900' → Alan Turing / Alonzo Church

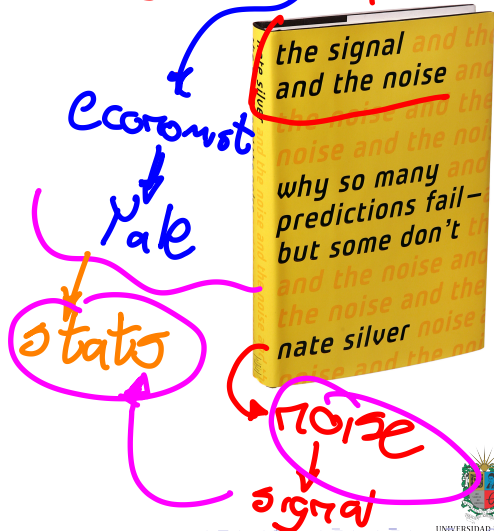


Synergy: Money Ball

→ Brad Pitt / Johan H. I.  
Manager The Athletics



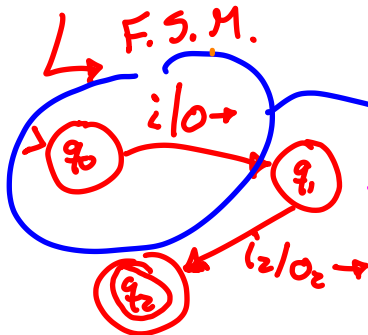
21-22



# Talking with Machines!

## Algorithm

F.S.M.



DAG

data pipelines → Apache Airflow

- **Alan Turing** proposed a hundred years ago an **Universal Machine**, capable of take any algorithm defined as a state machine, and process it in a binary language. → digital

- Forty years after, **Noam Chomsky** proposed the formal languages based on generative grammar. Here is were the high-level programming languages appear.

Input  
0101

~S.O.



# Talking with Machines!

$$\Sigma = \{a, b, c\}$$

$$G = \begin{cases} S \rightarrow aV_1 \\ V_1 \rightarrow bc/V_2 \\ \vdots \end{cases}$$

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60's

JVM

tree is binary  $\rightarrow$  CPU



# Programming Languages

- **Programming Languages** with more capabilities, easier comprehension had been created. Also, more people start to code into specific domain programming languages.

Natural Language

Demo time!

Java: hungry? bre: class;  
Python:

bre if hungry else class

- Andrej Karpathy, a leading figure at Tesla and now at OpenAI, said: Nowadays, Python is the most important programming language.

- SQL  
- Terraform (IAC)



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## Demo time!

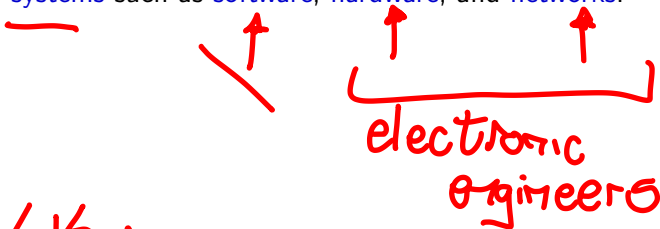
- **Andrej Karpathy**, a leading figure at **Tesla** and now at **OpenAI**, said: *Nowadays, English is the most important programming language.*

UM code ←  
 artificial agent  
 ↓  
 request code  
 [english]  
 #  
 code  
 vibing code



# Applications of Systems Engineering

- **Systems Engineering** can be applied to understand, analyze, and improve complex systems.
- **Systems Engineering** can be applied to design and manage complex systems such as organizations, cities, economies, and societies.
- **Systems Engineering** can be applied to design and manage complex systems such as software, hardware, and networks.



↳ dots  
↳ CTO → works



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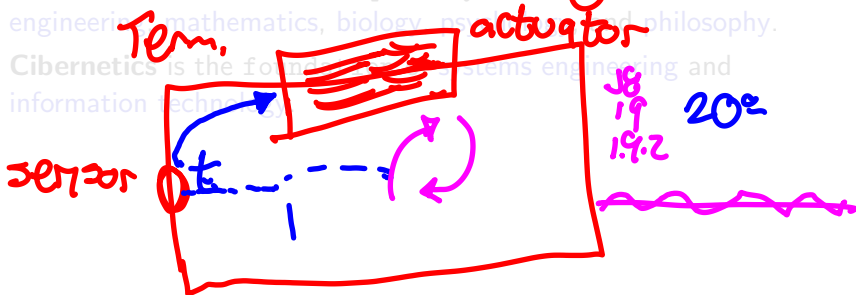


# Cibernetics

- Cibernetics is the study of systems, control, and communication in animals, machines, and organizations.

- Cibernetics is a transdisciplinary field that combines engineering, mathematics, biology, psychology, and philosophy.

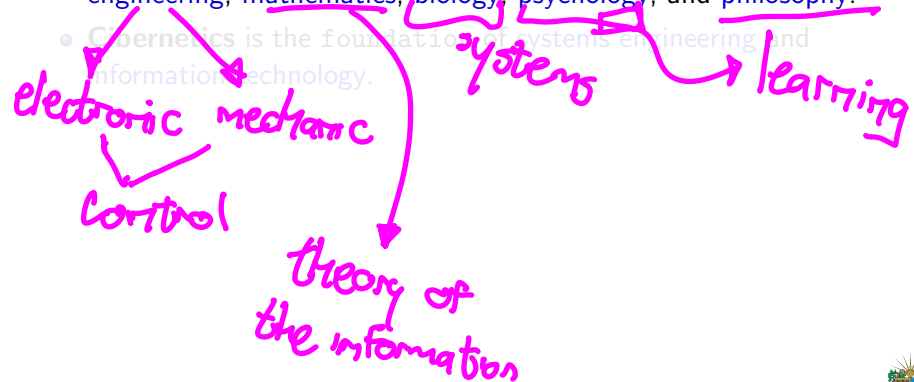
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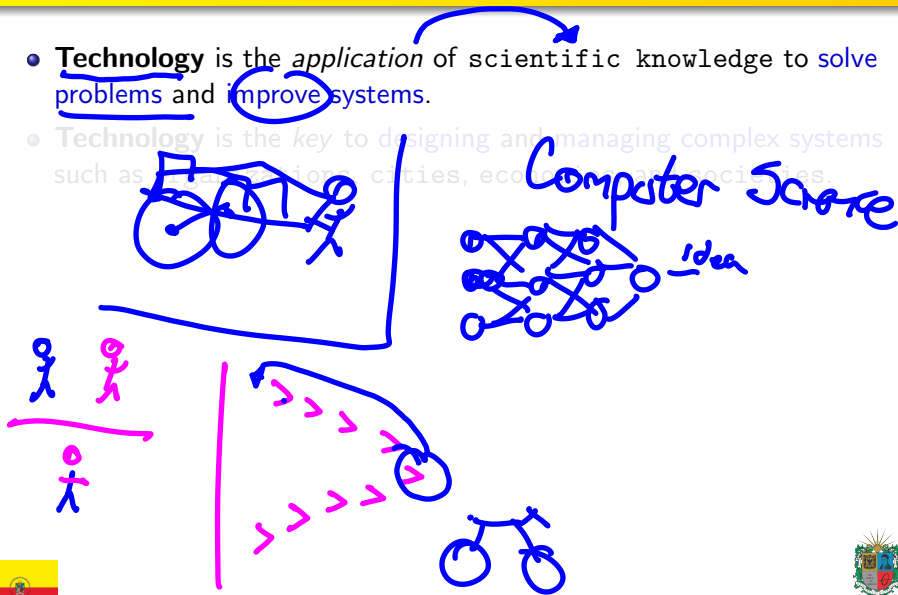
networks



# Technology

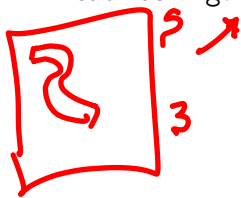
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- Technology is the key to designing and managing complex systems such as organizations, cities, ecosystems and societies.



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tech

theory  
of game

improve  
↓  
optimizations



## Technology

AI

1954

- **Artificial Intelligence** is a *field* of computer science that *studies* how to design and implement intelligent agents.

- Artificial Intelligence is the key to *designing* and *managing* complex systems such as smart-organizations, smart-cities, smart-economies, and smart-societies.

AI.MA.

→ solve problems like a human



# Technology: AI

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patterns  
recognition

information  
process  
speed

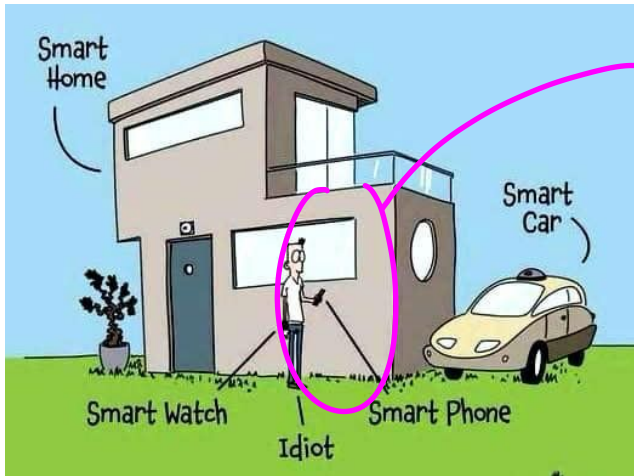
Chatbots

Human Academy  
Historical  
Figures

2016



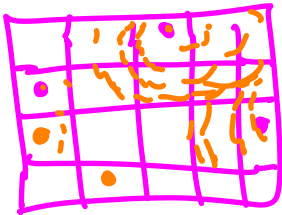
# Artificial Intelligence **not** as a System



# Cibernetics & Technology in Systems Context

- **Cibernetics** and **technology** are the foundation of systems engineering and information technology. ①
- **Cibernetics** and **technology** are the foundation of artificial intelligence and smart systems. ② ③

④

Sandpile

# Outline

■ Counting Sort → repeat

1 Basic Concepts

1 2 1 2 3 2 1 3 1 2 1 3 5 7

2 Human Activities

max 7 (n+1)

1 1 1 1 1 2 2 2 2 3 3 3 5 7

3 Cybernetics and Technology

0 1 2 3 4 5 6 7

4 Teams-Based Structure as a System



# Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are self-managed and cross-functional.
- Each team is **responsible** for a specific task or project and has the **authority** to make decisions and solve problems.
- **Team-based structure organizations** are flexible, agile, and innovative because they **empower employees** and **encourage collaboration**.



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# Teams as a System

- **Teams** are a **system** where collaboration (synergy) and communication are **key** to **success**.
- Defining and automating the **processes** and **procedures** within the **team** is a **challenge** to **improve team performance**.
- **Teams** are like **pieces** in a **puzzle**, where each **piece** has a **specific role** and **responsibility**.



# Soft Skills

- **Soft skills** are **personal attributes** that enable someone to **interact** effectively and harmoniously with other **people**.
- Typical Soft Skills:
  - **Communication** skills (verbal and written).
  - **Teamwork** and collaboration.
  - **Problem-solving** and critical thinking.
  - **Adaptability** and flexibility.
  - **Time management** and organization.
  - **Leadership** and management.
  - **Emotional intelligence**.
  - **Creativity** and innovation.
  - **Conflict resolution**.
  - **Networking** and relationship building.
  - **Customer service** and client management.



# Computer Analyst

## ● Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

## ● Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
- Providing support and training to end users.



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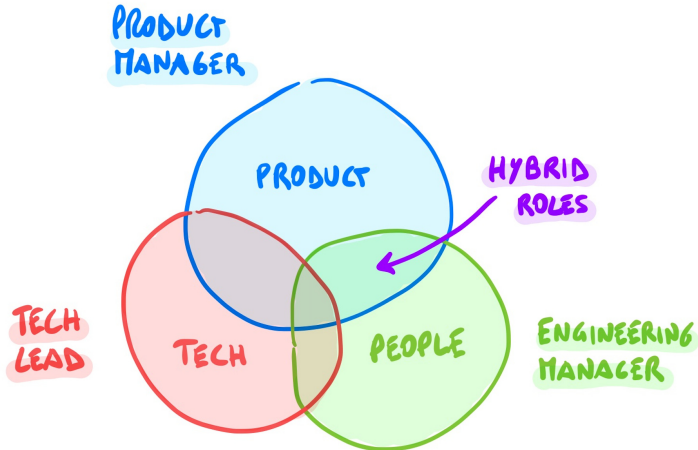
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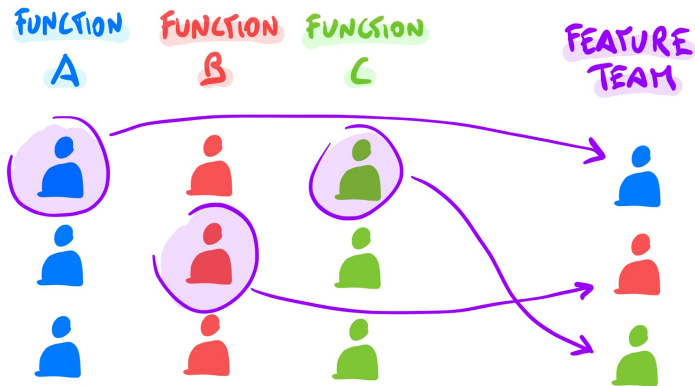
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# Tech Company Typical Structure



# Feature Teams



# What is to be a leader? I

- **Leading** a team is **not a role**. It is a decision, you could be a **leader** anytime and anywhere.
- **Teamwork culture** is pretty **important**. It creates habits, open communication, safety spaces for inclusion.
- **Psychological safety** is a **key** point to have an **effective** team. You could develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. Hierarchy exists by **status** and **power**.
- In a **hierarchy experts lead** to make **better decisions**. However, anyone must be **careful** to not **leave people behind**.



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# What is to be a leader? II

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- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just complete tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish — just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are vital.
- Making **ethical decisions** is **key**; it leads to taking the **right** and **better actions**.



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# What is to be a leader? III

- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



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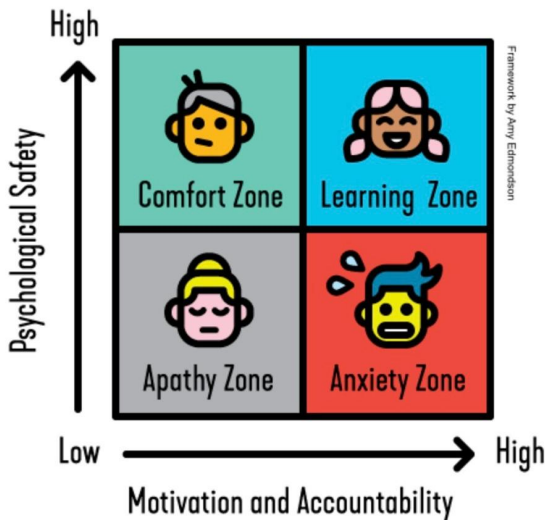


# What is to be a leader? III

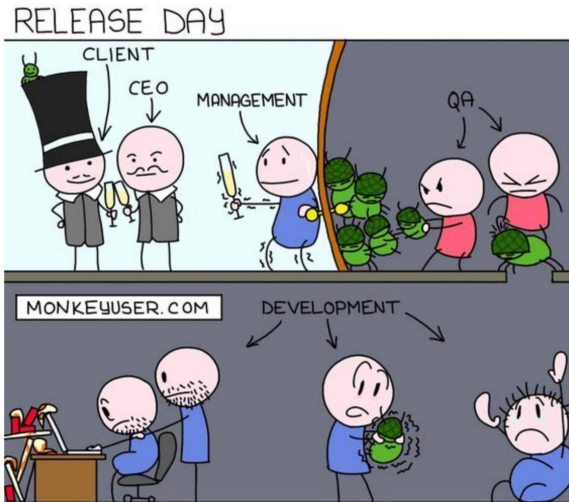
- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To **develop as a leader**, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



# Working Zones



# Real World!



# Outline

- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



# Thanks!

## Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

